A Brief Idea About Three-Phase Inverters

Presented by

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Recap...

A Three Phase inverter can be realized by THREE SINGLE PHASE (FULL-)BRIDGE INVERTERS, connected in parallel and driven by 120^0(elect.) apart shown in fig. 1

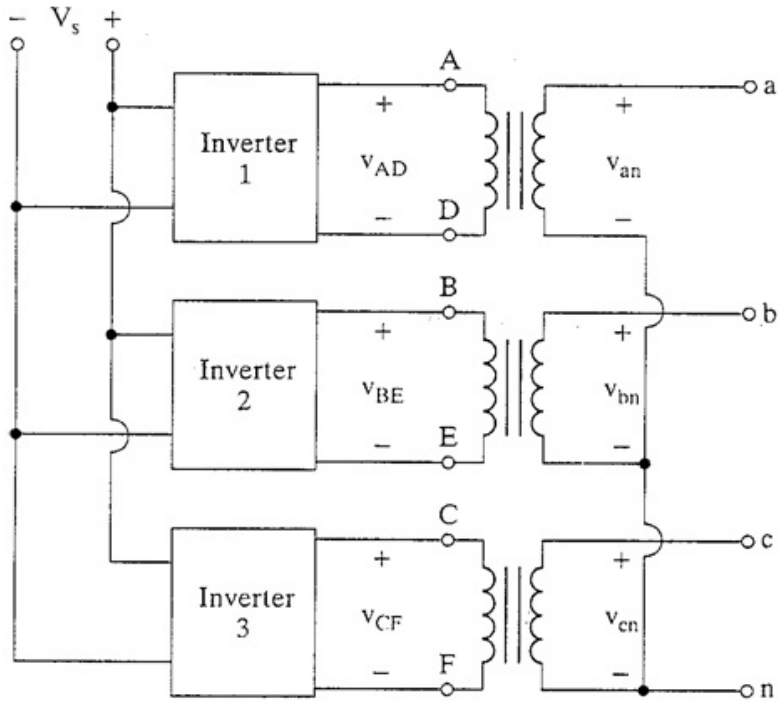


Figure 1: Three Phase Inverter Realized By Single Phase Inverters

Recap...

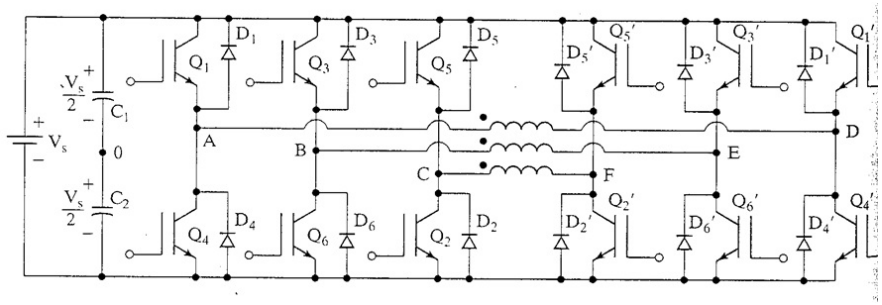


Figure 2: Three single-phase full bridge inverters

Above circuit (in fig. 2) has

- 12 transistors
- 12 diodes
- 3 transformers

Could it be simpler? ? ? is there any "Alternative" ???

Recap...

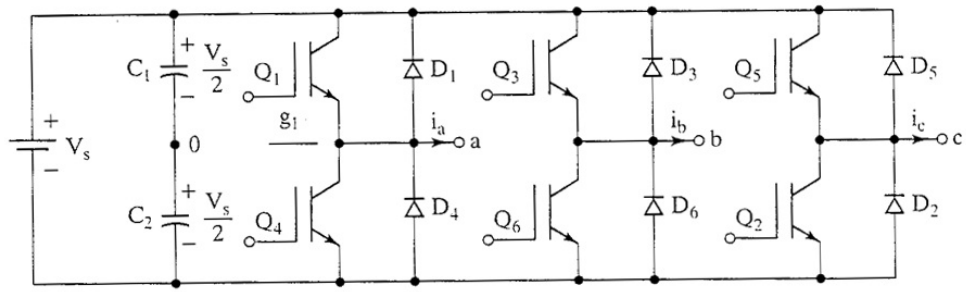


Figure 3: Three Phase Inverter Circuit

This above circuit (in fig. 3) has

- 6 transistors
- 6 diodes

This circuit can be operated as follows

- 120^0 mode
- 180^0 mode

180^0 Conduction:

In this mode of operation any three switches[1] will be ON at a time
The logic table is shown in fig. 4

TABLE 6.2 Switch States for Three-Phase Voltage-Source Inverter (VSI)

State	State No.	Switch States	v_{ab}	v_{bc}	v_{ca}
S_1, S_2, and S_6 are on and S_4, S_5, and S_3 are off	1	100	V_S	0	$-V_S$
S_2, S_3, and S_1 are on and S_5, S_6, and S_4 are off	2	110	0	V_S	$-V_S$
S_3, S_4, and S_2 are on and S_6, S_1, and S_5 are off	3	010	$-V_S$	V_S	0
S_4, S_5, and S_3 are on and S_1, S_2, and S_6 are off	4	011	$-V_S$	0	V_S
S_5, S_6, and S_4 are on and S_2, S_3, and S_1 are off	5	001	0	$-V_S$	V_S
S_6, S_1, and S_5 are on and S_3, S_4, and S_2 are off	6	101	V_S	$-V_S$	0
S_1, S_3, and S_5 are on and S_4, S_6, and S_2 are off	7	111	0	0	0
S_4, S_6, and S_2 are on and S_1, S_3, and S_5 are off	8	000	0	0	0

Figure 4: Summary Table for 180^0 mode

[1] must not 2 switches of the same leg

Recap...

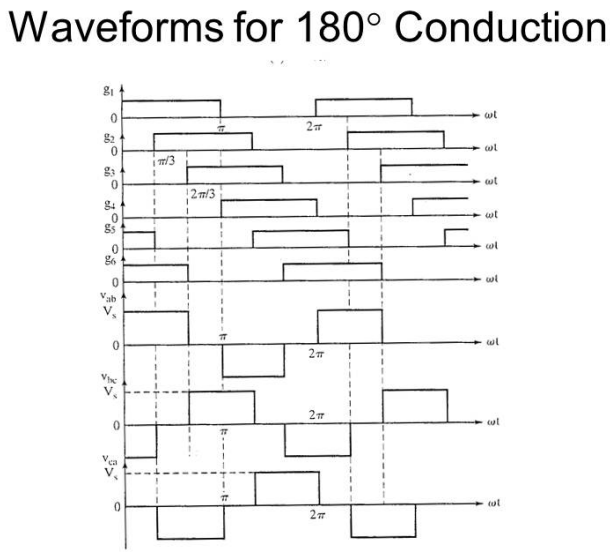


Figure 5: Gate Pulses & L-L voltages

Recap..Mode 1: SW6,SW1, SW5 are ON:

Summary Table

TABLE 6.2 Switch States for Three-Phase Voltage-Source Inverter (VSI)

State	State No.	Switch States	v_{ab}	v_{bc}	v_{ca}
S_1, S_2, and S_6 are on and S_4, S_5, and S_3 are off	1	100	V_S	0	$-V_S$
S_2, S_3, and S_1 are on and S_5, S_6, and S_4 are off	2	110	0	V_S	$-V_S$
S_3, S_4, and S_2 are on and S_6, S_1, and S_5 are off	3	010	$-V_S$	V_S	0
S_4, S_5, and S_3 are on and S_1, S_2, and S_6 are off	4	011	$-V_S$	0	V_S
S_5, S_6, and S_4 are on and S_2, S_3, and S_1 are off	5	001	0	$-V_S$	V_S
S_6, S_1, and S_5 are on and S_3, S_4, and S_2 are off	6	101	V_S	$-V_S$	0
S_1, S_3, and S_5 are on and S_4, S_6, and S_2 are off	7	111	0	0	0
S_4, S_6, and S_2 are on and S_1, S_3, and S_5 are off	8	000	0	0	0

Figure 6: MODE 1 logic table

Recap...

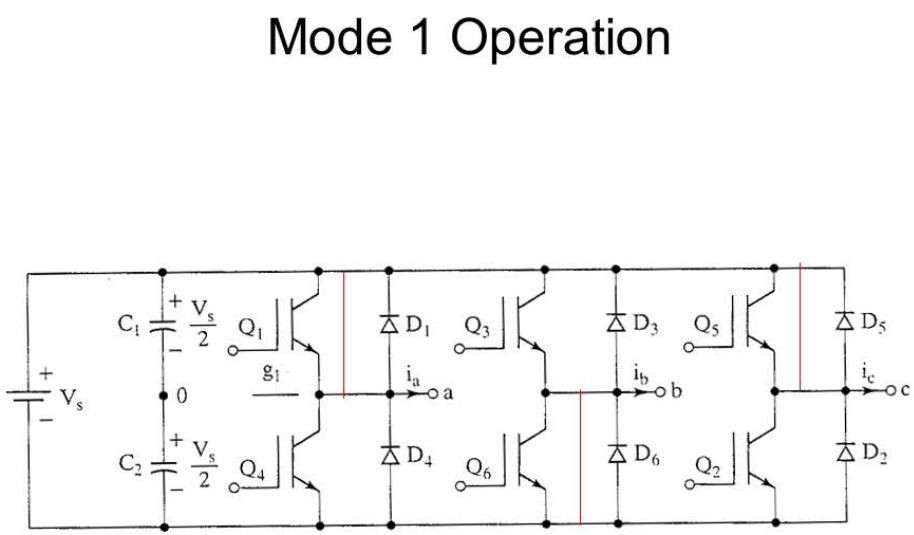
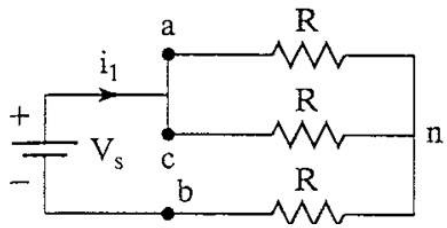


Figure 7: Mode 1 : circuit condition

Recap...

Mode 1 Operation

$$0 \le \omega t \le \frac{\pi}{3}$$



Q_1, Q_5, Q_6 conduct

$$R_{eq} = R + \frac{R}{2} = \frac{3R}{2}$$

$$i_1 = \frac{V_s}{R_{eq}} = \frac{2V_s}{3R}$$

$$v_{an} = v_{cn} = \frac{i_1 R}{2} = \frac{V_s}{3}$$

$$v_{bn} = -i_1 R = \frac{-2V_s}{3}$$

Figure 8: Mode 1 : Calculation

Recap...

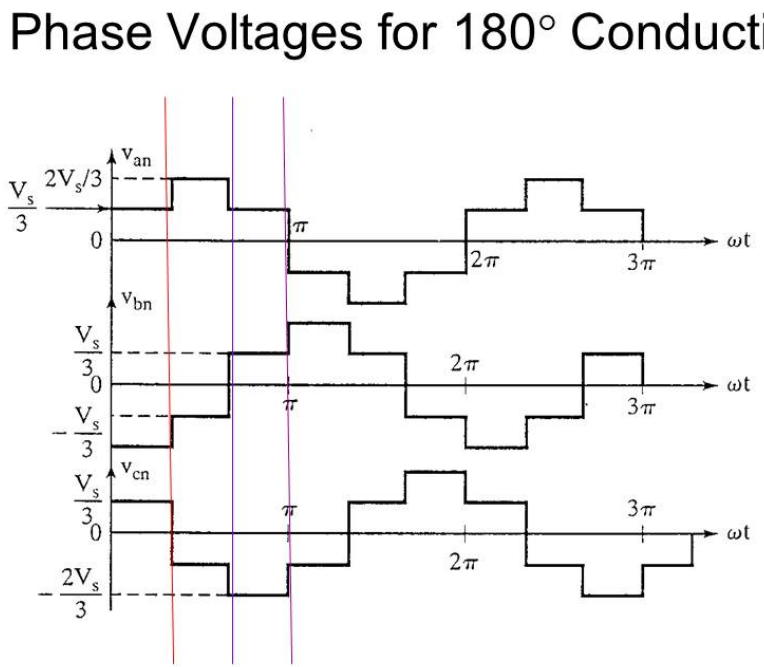


Figure 9: Mode 1: Phase Voltage waveform

Recap...

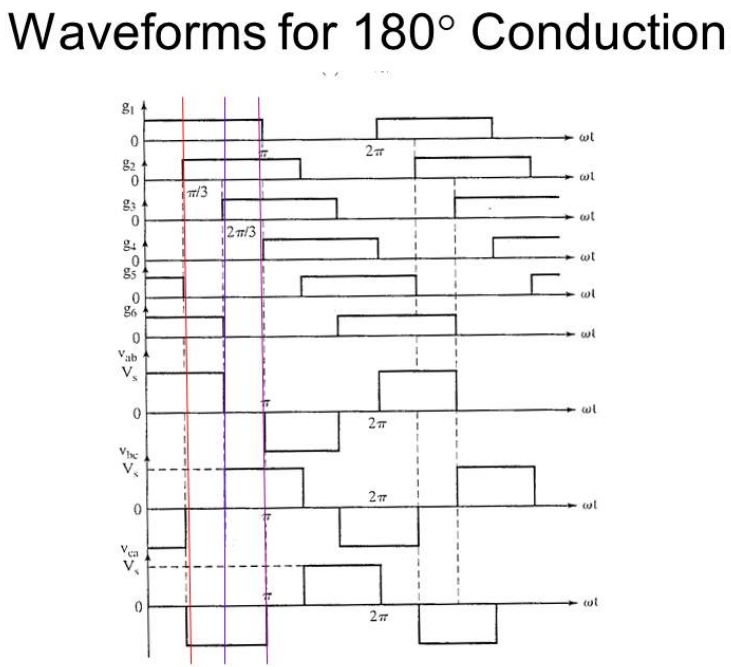


Figure 10: Mode 1: Gate pulse and line voltage waveform

Recap...Mode 2:SW6,SW1, SW2 are ON:

Summary Table

TABLE 6.2 Switch States for Three-Phase Voltage-Source Inverter (VSI)

State	State No.	Switch States	v_{ab}	v_{bc}	v_{ca}
S_1, S_2, and S_6 are on and S_4, S_5, and S_3 are off	1	100	V_S	0	$-V_S$
S_2, S_3, and S_1 are on and S_5, S_6, and S_4 are off	2	110	0	V_S	$-V_S$
S_3, S_4, and S_2 are on and S_6, S_1, and S_5 are off	3	010	$-V_S$	V_S	0
S_4, S_5, and S_3 are on and S_1, S_2, and S_6 are off	4	011	$-V_S$	0	V_S
S_5, S_6, and S_4 are on and S_2, S_3, and S_1 are off	5	001	0	$-V_S$	V_S
S_6, S_1, and S_5 are on and S_3, S_4, and S_2 are off	6	101	V_S	$-V_S$	0
S_1, S_3, and S_5 are on and S_4, S_6, and S_2 are off	7	111	0	0	0
S_4, S_6, and S_2 are on and S_1, S_3, and S_5 are off	8	000	0	0	0

Figure 11: Mode 2: Logic Table

Recap...

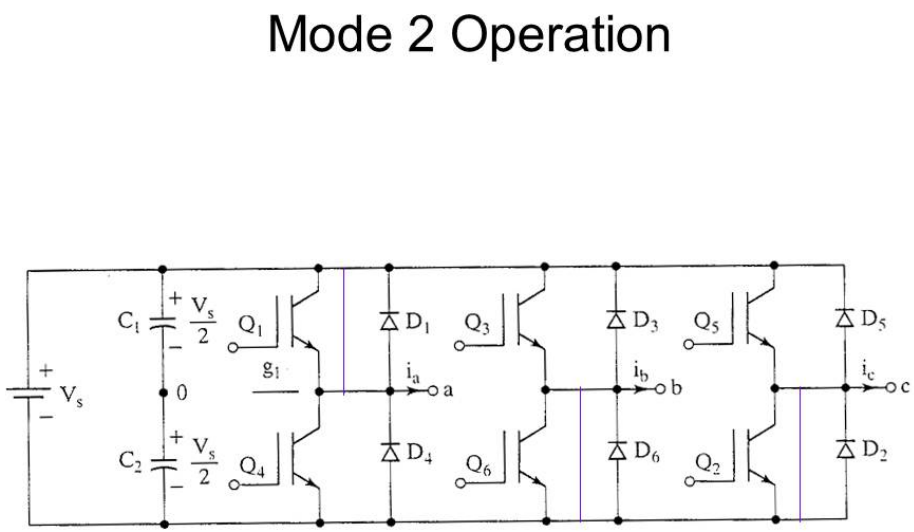
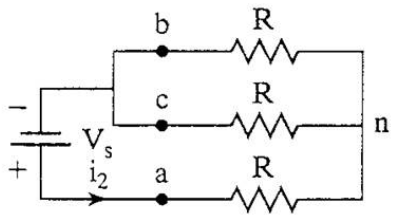


Figure 12: Mode 2: circuit condition

Recap...

Mode 2 Operation

$$\frac{\pi}{3} \le \omega t \le \frac{2\pi}{3}$$



Q_1, Q_2, Q_6 conduct

$$R_{eq} = R + \frac{R}{2} = \frac{3R}{2}$$

$$i_2 = \frac{V_s}{R_{eq}} = \frac{2V_s}{3R}$$

$$v_{an} = i_2 R = \frac{2V_s}{3}$$

$$v_{bn} = v_{cn} = \frac{-i_2 R}{2} = \frac{-V_s}{3}$$

Figure 13: Mode 2: Calculation

Recap...

Fourier Series for Line-to-Line Voltages

$$v_{ab} = \frac{a_o}{2} + \sum_{n=1}^{\infty} (a_n \cos(n\omega t) + b_n \sin(n\omega t))$$

$$b_n = \frac{1}{\pi}\left[\int_{\frac{-5\pi}{6}}^{\frac{5\pi}{6}} -V_s d(\omega t) + \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} V_s d(\omega t)\right]$$

$$b_n = \frac{4V_s}{n\pi}\sin(\frac{n\pi}{2})\sin(\frac{n\pi}{3})$$

$$\boxed{v_{ab} = \sum_{n=1,3,5,\ldots}^{\infty} \frac{4V_s}{n\pi}\sin\frac{n\pi}{3}\sin n(\omega t + \frac{\pi}{6})}$$

Recap...

For the other Line-to-Line Voltages

$$v_{bc} = \sum_{n=1,3,5,\ldots}^{\infty} \frac{4V_s}{n\pi} \sin\frac{n\pi}{3} \sin n(\omega t - \frac{\pi}{2})$$

$$v_{ca} = \sum_{n=1,3,5,\ldots}^{\infty} \frac{4V_s}{n\pi} \sin\frac{n\pi}{3} \sin n(\omega t - \frac{7\pi}{6})$$

Recap...

Line-to-Line rms Voltage

$$V_L = \left[\frac{2}{2\pi} \int_0^{\frac{2\pi}{3}} V_s^2 \, d(\omega t) \right]^{\frac{1}{2}}$$

$$V_L = \sqrt{\frac{2}{3}} V_s = 0.8165 V_s$$

Recap...

rms value of the nth Component

$$V_{Ln} = \frac{4V_s}{\sqrt{2}n\pi}\sin\frac{n\pi}{3}$$

$n = 1$ Fundamental Component

$$V_{L1} = \frac{4V_s \sin 60°}{\sqrt{2}\pi} = 0.7797V_s$$

Recap...

Line-to-Neutral Voltages

$$V_p = \frac{V_L}{\sqrt{3}} = \frac{\sqrt{2}V_s}{3} = 0.4714V_s$$

Line Current for an RL load

$$i_a = \sum_{n=1,3,5,\ldots}^{\infty} \left[\frac{4V_s}{\sqrt{3}\left[n\pi\sqrt{R^2 + (n\omega L)^2} \right]} \sin\frac{n\pi}{3} \right] \sin(n\omega t - \theta_n)$$

$$\theta_n = \tan^{-1}\left(\frac{n\omega L}{R} \right)$$

Recap...

DC Supply Current

$$v_s i_s = v_{ab}(t) i_a(t) + v_{bc}(t) i_b(t) + v_{ca}(t) i_c(t)$$

....

$$I_s = 3 \frac{V_{o1}}{V_s} I_o \cos(\theta_1)$$

$$I_s = \sqrt{3} \frac{V_{o1}}{V_s} I_L \cos(\theta_1)$$

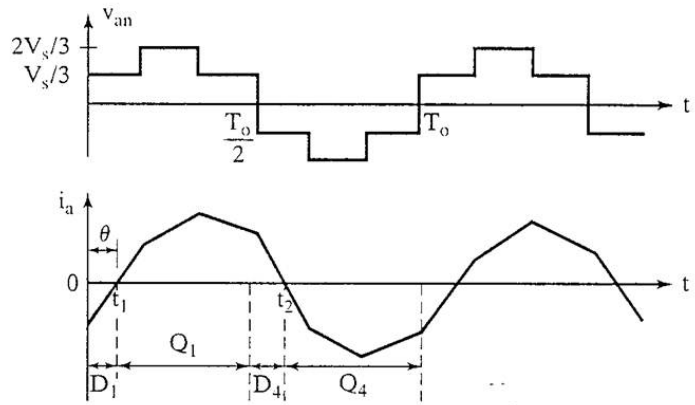
$I_L = \sqrt{3} I_o$ is the rms load line current

V_{o1}= fundamental rms output line voltage

I_o is the rms load phase current

Θ_1 = the load impedance angle at the fundamental frequency

Three-Phase Inverter with RL Load



120^0 conduction

- each S/W will conduct for 120^0 (elect.).
- At any instant **ONLY TWO** S/Ws will be on
- After each 60^0 a old S/W is replaced by new S/W

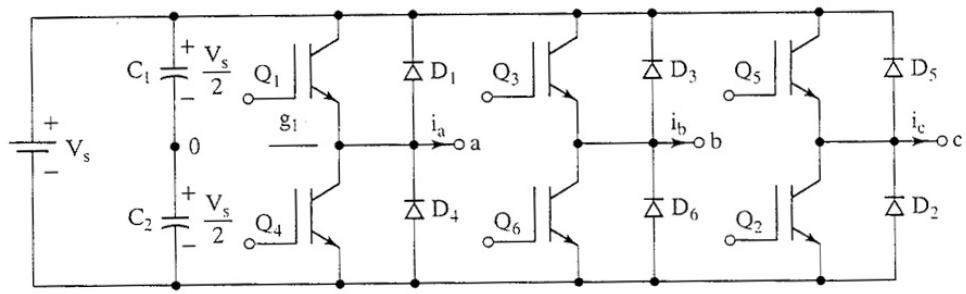


Figure 14: Three Phase Inverter Circuit

Switching sequence and gate pulses for 120^0 Conduction:

$$6,1 \rightarrow 1,2 \rightarrow 2,3 \rightarrow 3,4 \rightarrow 4,5 \rightarrow 5,6 \rightarrow 6,1$$

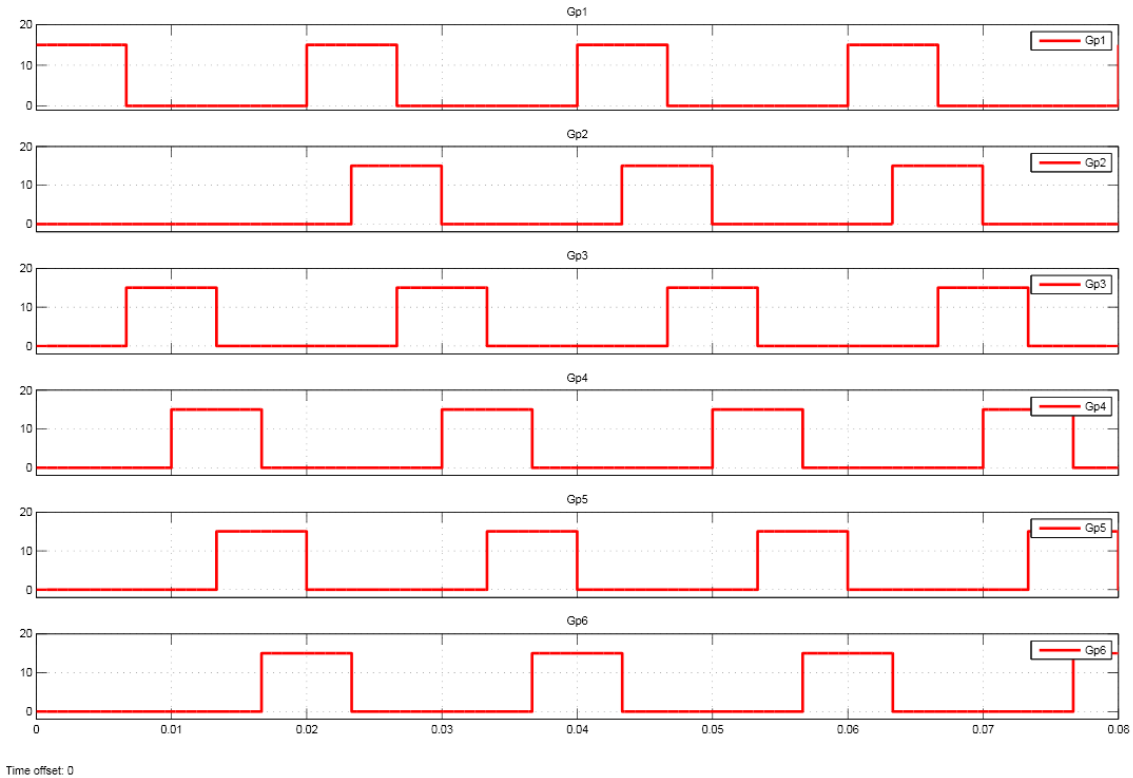


Figure 15: Gate Pulses for 120^0 conduction

Mode 1: SW 6,1 ON for $0 \leq \omega t \leq \frac{\pi}{3}$

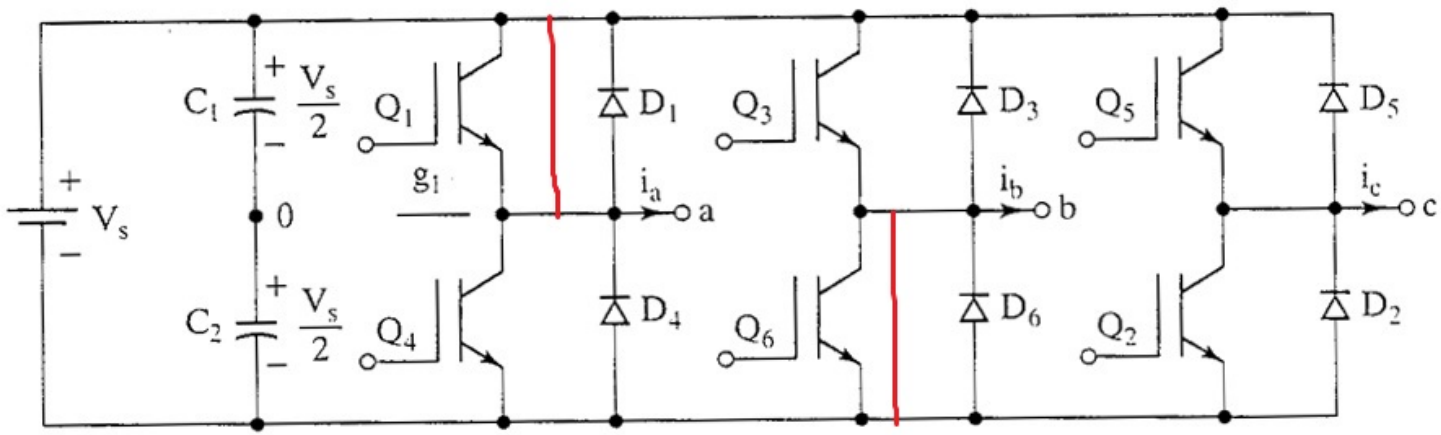


Figure 16: Mode 1: circuit condition

The equivalent circuit of fig:16 is

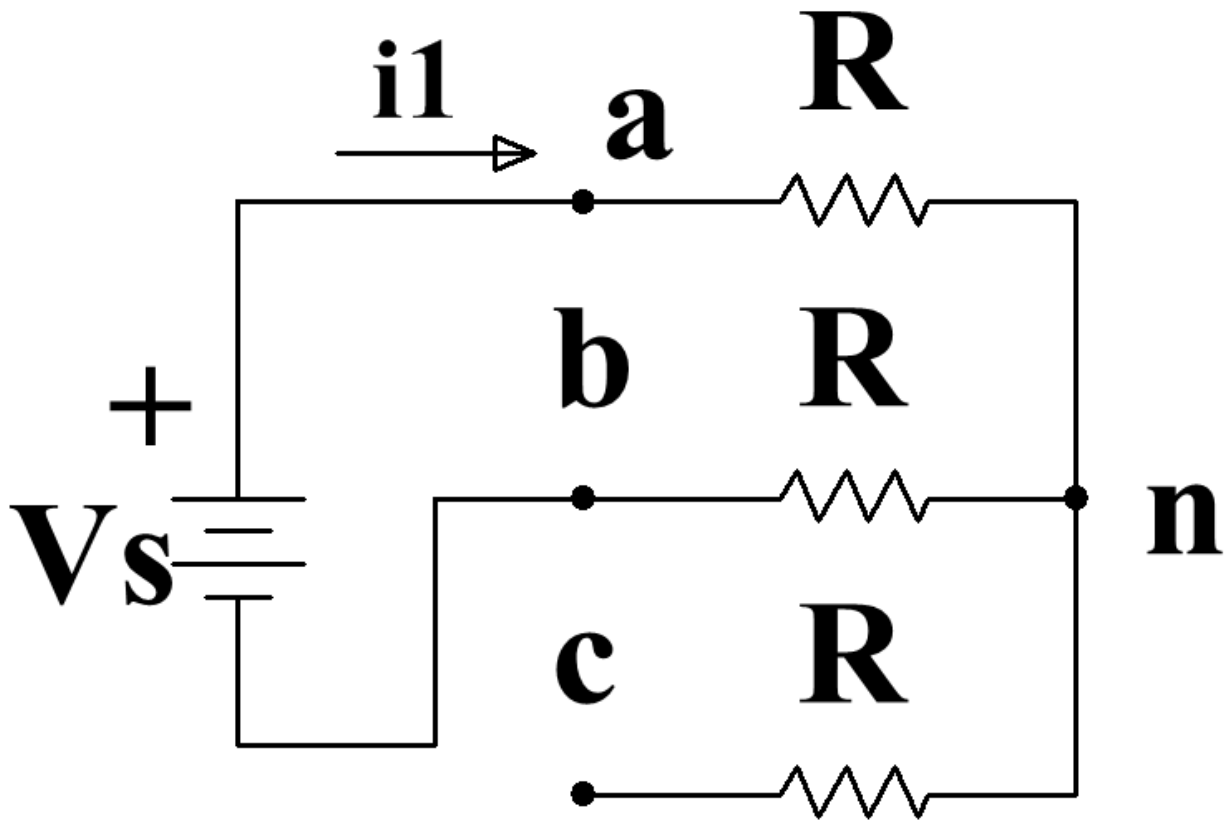


Figure 17: Equivalent circuit for mode 1

Hence, $v_{an} = \frac{V_s}{2}$; $v_{bn} = -\frac{V_s}{2}$; $v_{cn} = 0\star$

Mode 2: SW 1,2 ON for $\frac{\pi}{3} \leq \omega t \leq \frac{2\pi}{3}$

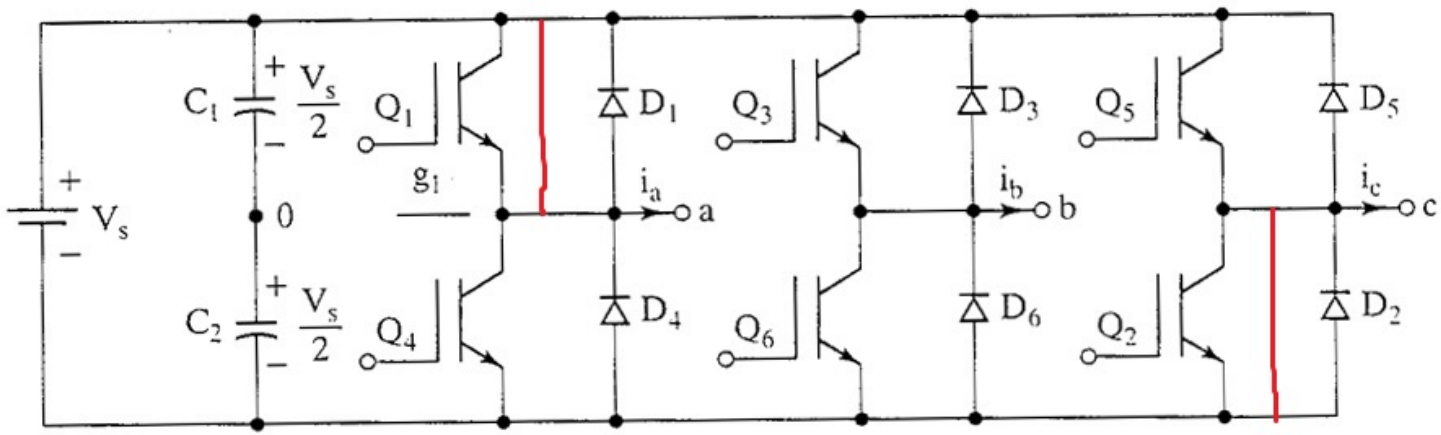


Figure 18: Mode 2: circuit condition

The equivalent circuit of fig:18 is

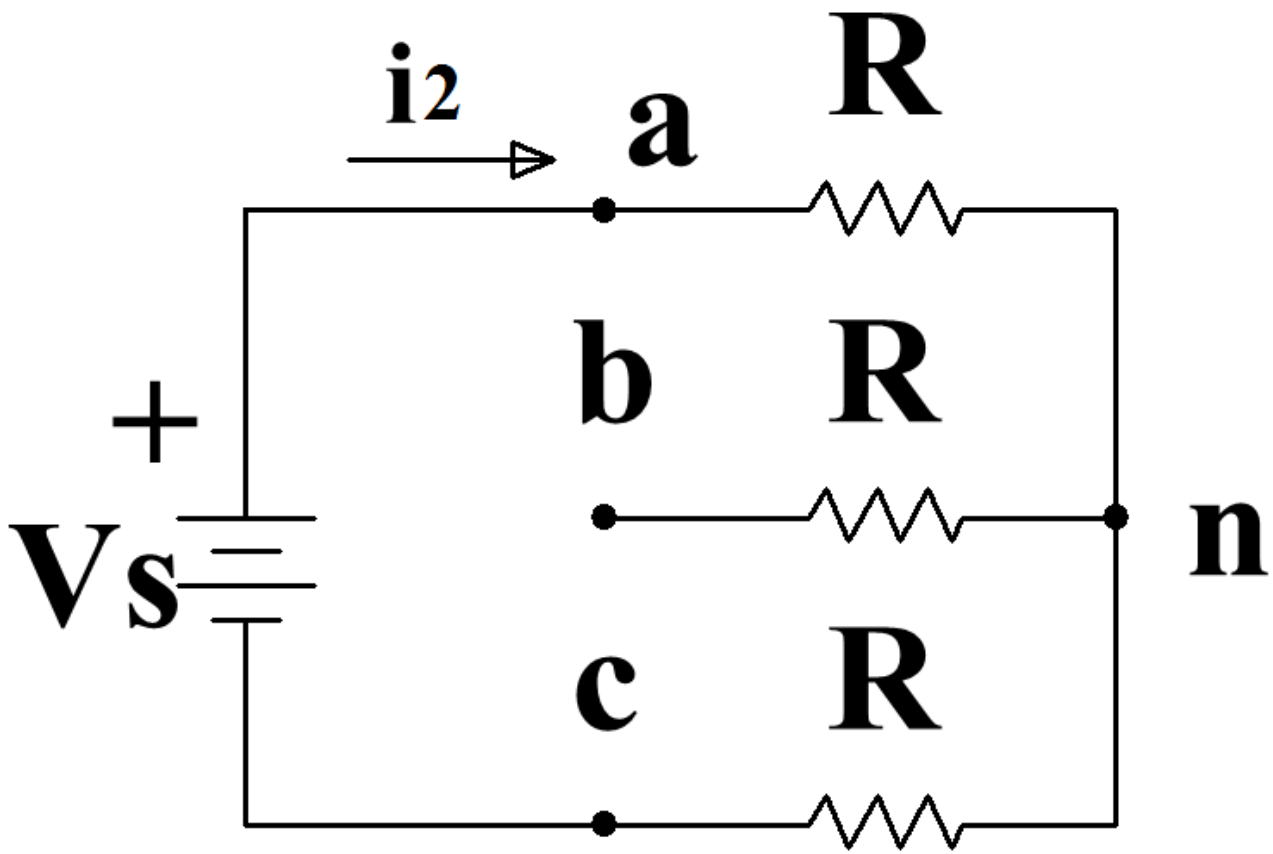


Figure 19: Equivalent circuit for mode 2

Hence, $v_{an} = \frac{V_s}{2}$; $v_{bn} = 0\star$; $v_{cn} = -\frac{V_s}{2}$

Mode 3: SW ON for $\frac{2\pi}{3} \leq \omega t \leq \frac{3\pi}{3}$

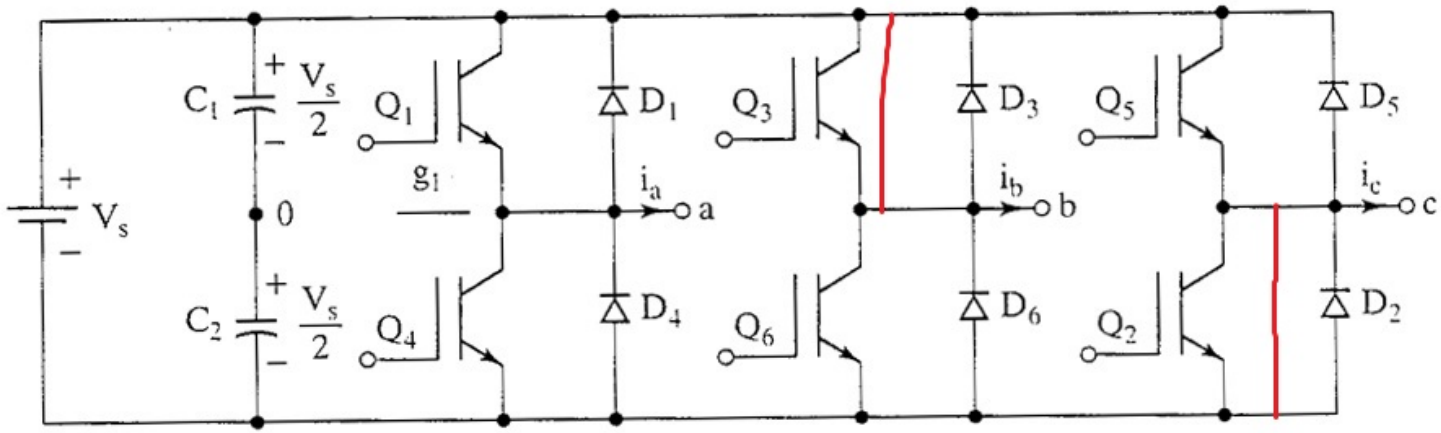


Figure 20: Mode 3: circuit condition

The equivalent circuit of fig:20 is

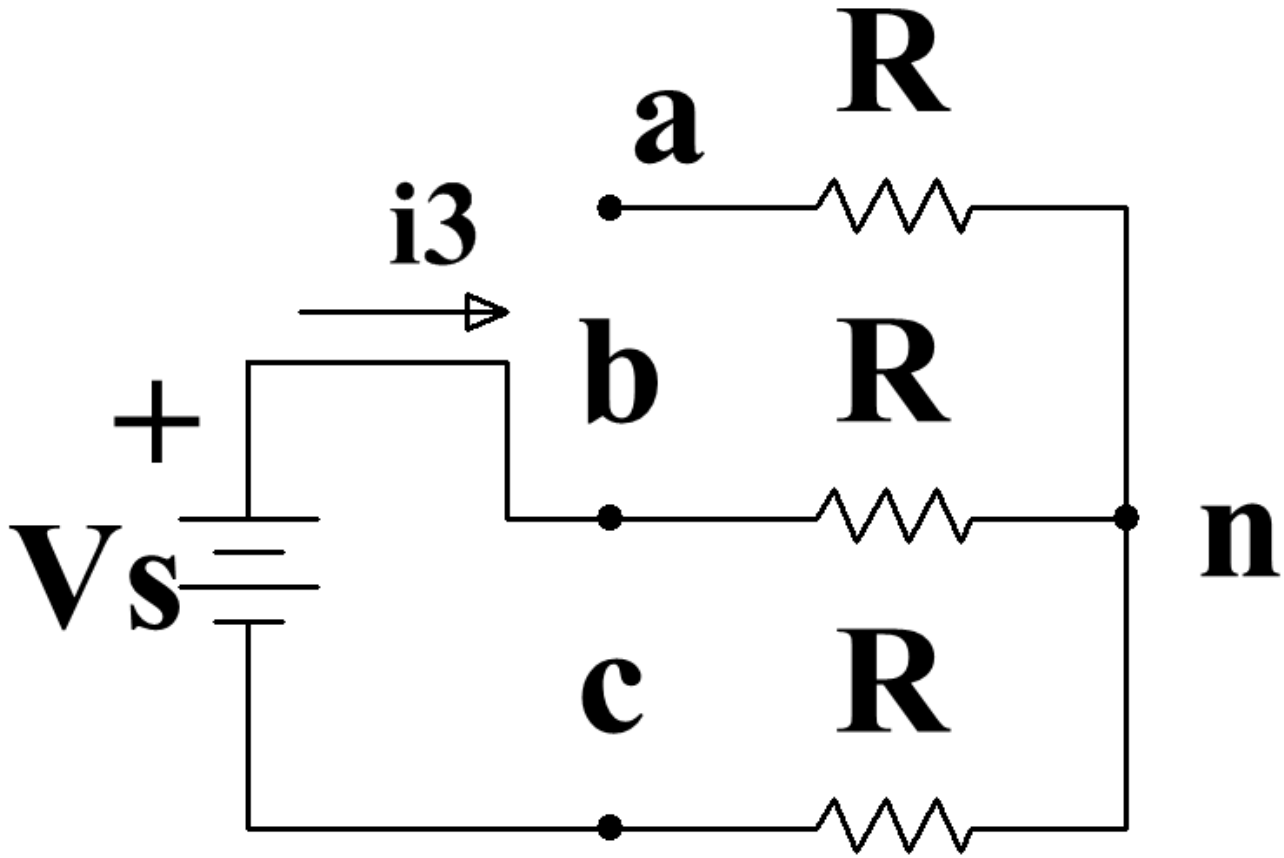


Figure 21: Equivalent circuit for mode 3

Hence, $v_{an} = 0\star$; $v_{bn} = \frac{V_s}{2}$; $v_{cn} = -\frac{V_s}{2}$

phase and Line-Line voltage:

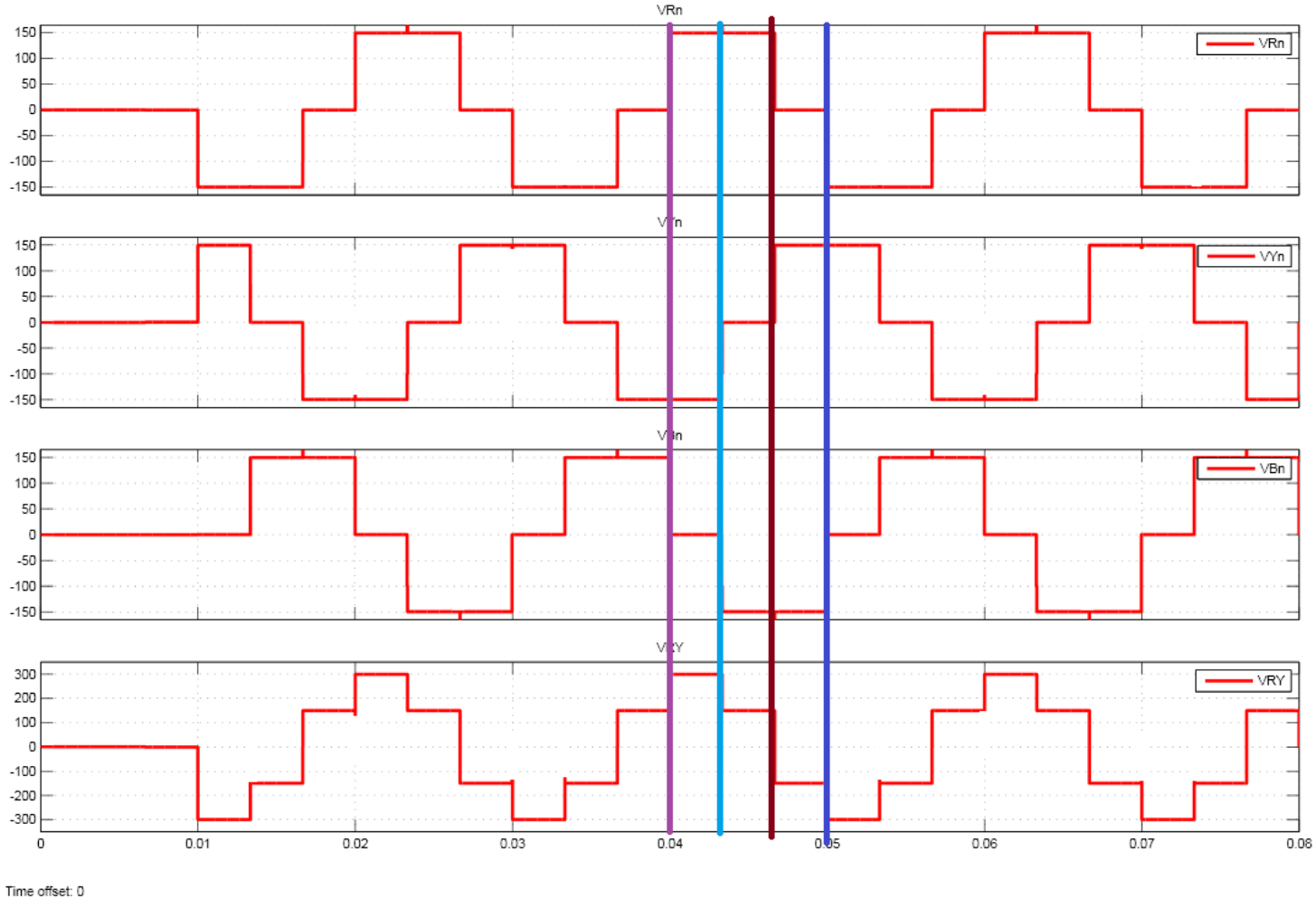


Figure 22: Phase and Line-Line Voltage of 120^0 Conduction

The line to neutral voltages(in fig:22) can be expressed as in Fourier's series,

$$v_{kn} = \sum_{n=1,3,5,\ldots}^{\infty} \frac{2V_s}{n\pi} \sin\frac{n\pi}{3} \sin n(\omega t + \frac{\pi}{6} + \phi_k) \qquad (1)$$

k	ϕ_k (in rad)
phase-R	0
phase-Y	$-\frac{2\pi}{3}$
phase-B	$-\frac{4\pi}{3}$

The line to line voltages(in fig:22) can be expressed as in Fourier's series,

$$v_{kn} = \sum_{n=1,3,5,\ldots}^{\infty} \frac{2\sqrt{3}V_s}{n\pi} \sin\frac{n\pi}{3} \sin n(\omega t + \frac{\pi}{6} + \phi_k) \qquad (2)$$

k	ϕ_k (in rad)
phase-R	$0 + \frac{\pi}{6}$
phase-Y	$-\frac{2\pi}{3} + \frac{\pi}{6}$
phase-B	$-\frac{4\pi}{3} + \frac{\pi}{6}$

Comparision between 120^0 and 180^0:

Parameters	180^0	120^0
no. of s/w conductiong at a time	3	2
each s/w condt. for	180^0 (elect)	120^0 (elect)
control syst. point of views	good control	not good
V_{L1}	$0.7798V_s$	$0.6753V_s$
THD	31%	31%

MATLAB model for 180^0 3-ph Inverter:

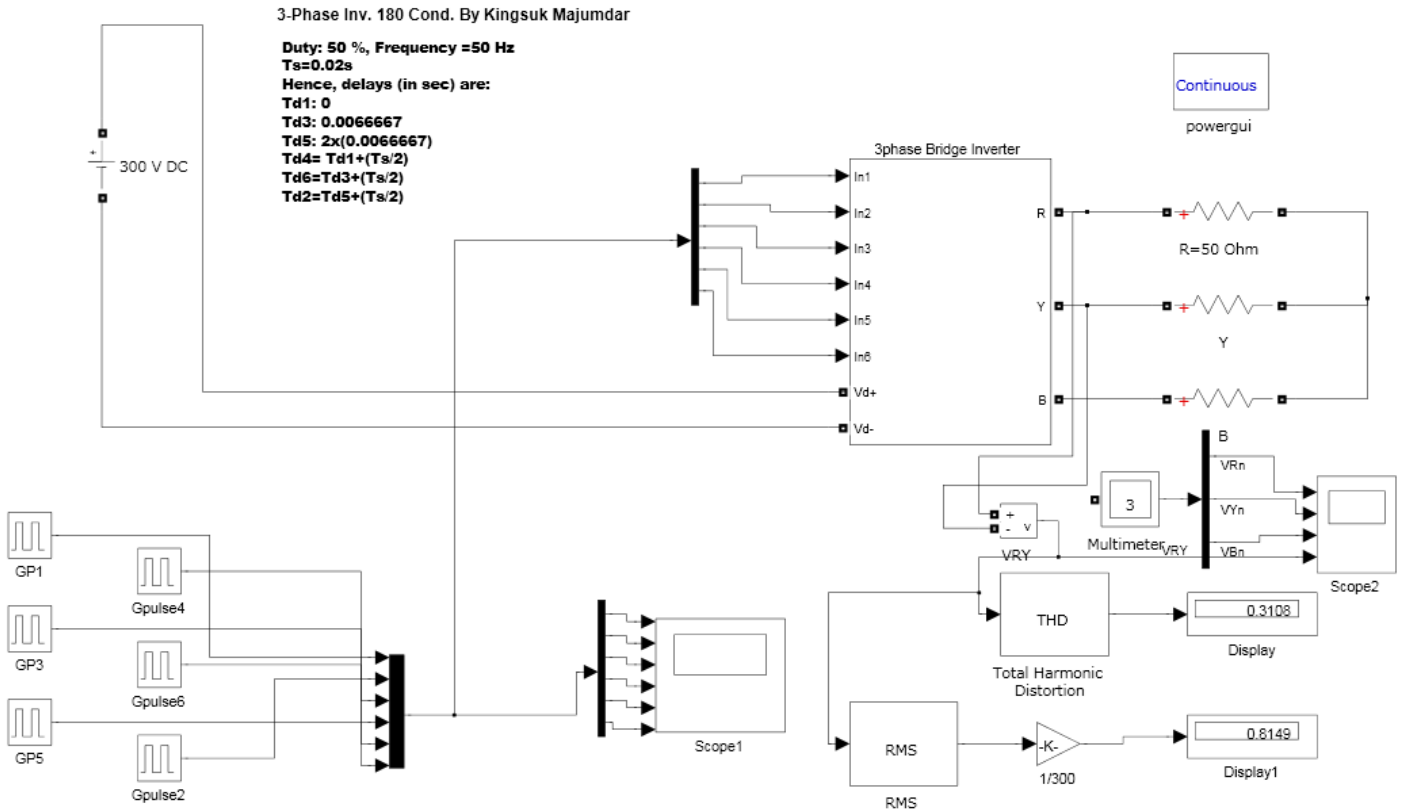


Figure 23: MATLAB Model of 180^0 3ph-Inverter

MATLAB model for 120^0 3-ph Inverter:

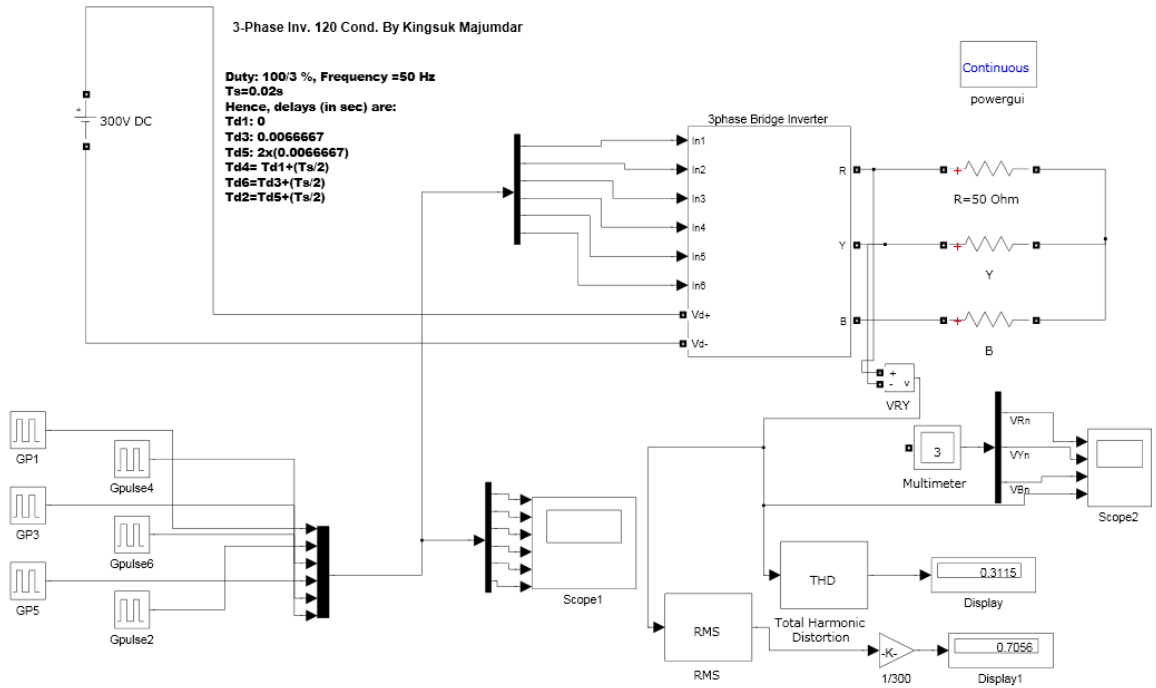


Figure 24: MATLAB Model of 120^0 3ph-Inverter

Thank You...